

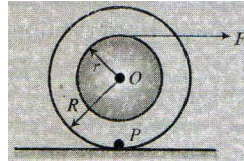
$$mg \cos 60^\circ (l/4) = \left[\frac{ml^2}{12} + m \left(\frac{l}{4} \right)^2 \right] \alpha \quad \therefore \quad \alpha = \frac{6g}{7l}$$

(e) Now, N will have different value

$$\text{About C, } N \frac{l}{4} = I_c \alpha = \frac{ml^2}{12} \times \frac{6g}{7l} = \frac{mgl}{14}$$

$$\therefore \quad N = 2mg/7$$

5. A cotton reel of mass m and moment of inertia I is kept at rest on a smooth horizontal surface. The reel has inner and outer radius r and R respectively. A horizontal force F starts acting as shown in fig. Find the



- acceleration of the centre of mass of reel.
- angular acceleration of the reel
- net acceleration of point of contact.

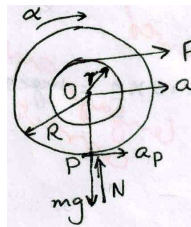
Solution The only force acting in horizontal direction is F .

$$\text{From FBD, } ma = F \quad \therefore \quad a = F/m$$

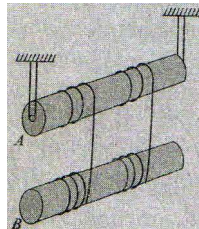
$$\text{About O, } I\alpha = Fr \quad \therefore \quad \alpha = Fr/I$$

$$a_0 - a_p = R\alpha \quad \Rightarrow \quad a - a_p = R(Fr/I)$$

$$\therefore \quad a_p = a - FRr/I$$



6. The arrangement shown in fig, consists of two identical, uniform, solid cylinders, each of mass m , on which two light threads are wound symmetrically. Find the tensions of each thread in the process of motion. The friction in the axle of the upper cylinder is assumed to be absent.



Solution For upper pulley, $2Tr = \frac{mr^2}{2} \alpha_1$

$$\Rightarrow \alpha_1 = 4T/mr \quad \Rightarrow \quad a_A = \alpha_1 r = 4T/m$$

For lower pulley, $a_B = a_A = 4T/m$

$$ma_C = mg - 2T \quad \Rightarrow \quad a_C = g - 2T/m$$

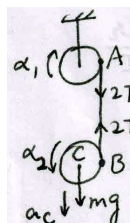
Taking torque about C,

$$2Tr = \frac{mr^2}{2} \alpha_2 \quad \Rightarrow \quad \alpha_2 = \frac{4T}{mr}$$

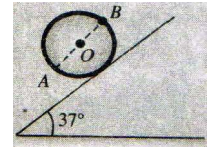
$$a_C - a_B = \alpha_2 r$$

$$\Rightarrow \left(g - \frac{2T}{m} \right) - \left(\frac{4T}{m} \right) = \frac{4T}{m}$$

$$\therefore \quad T = \frac{mg}{10}$$



7. A uniform solid sphere of mass 1 kg and radius 10 cm is kept stationary on a rough inclined plane by fixing a highly dense particle at B. Inclination of plane is 37° with horizontal and AB is the diameter of the sphere which is parallel to the plane, as shown in figure. Calculate



(a) mass of the particle at B

(b) minimum required coefficient of friction between sphere and plane to keep sphere in equilibrium.

Solution Since the system is in equilibrium, net force and net torque is zero. Along the incline,

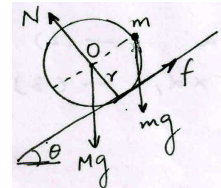
$$(M + m)g \sin \theta = f \quad \dots(1)$$

Perpendicular to incline,

$$N = (M + m)g \cos \theta \quad \dots(2)$$

Taking torque about O, we have

$$fr = mgr \cos \theta \quad \Rightarrow \quad f = mg \cos \theta \quad \dots(3)$$



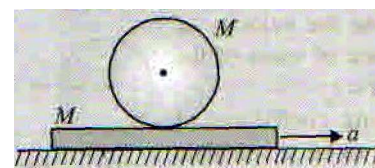
(a) From (1) and (3), $(M + m)g \sin \theta = mg \cos \theta$

$$\therefore \quad m = \frac{M \tan \theta}{1 - \tan \theta} = \frac{1 \times 0.75}{1 - 0.75} = 3 \text{ kg}$$

(b) $f \leq \mu N$, $\Rightarrow mg \cos \theta \leq \mu (M + m)g \cos \theta$

$$\Rightarrow \mu \geq \frac{m}{M + m} = \frac{3}{3 + 1} \quad \therefore \quad \mu \geq 0.75$$

8. A uniform cylinder (mass M) of radius R is kept on an accelerating platform (mass M) as shown in fig. If the cylinder rolls without slipping on the platform, determine the magnitude of acceleration of the centre of mass of the cylinder. Assuming the coefficient of friction $\mu = 0.40$, determine the maximum acceleration the platform may have without slipping between the cylinder and the platform.



Solution From FBD, $f = Ma_c \quad \Rightarrow \quad a_c = f/M$

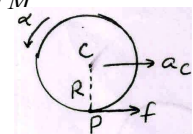
$$fR = I_C \alpha = \frac{MR^2}{2} \alpha \quad \Rightarrow \quad \alpha = \frac{2f}{MR}$$

For no slipping at point P, $a_p - a_c = \alpha R$

$$\Rightarrow a - \frac{f}{M} = \frac{2f}{MR} \times R \quad \Rightarrow \quad f = Ma/3$$

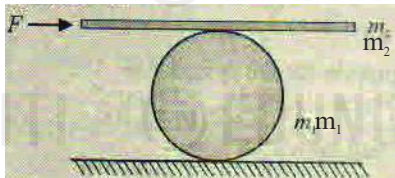
$$\therefore \quad a_c = \frac{f}{M} = \frac{a}{3}$$

$$f \leq \mu N \quad \Rightarrow \quad \frac{Ma}{3} \leq \mu Mg$$



$$\therefore a_{\max} = 3\mu g$$

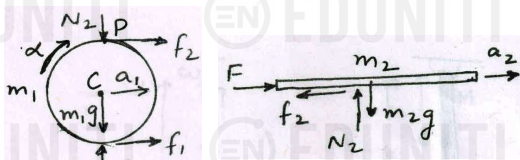
9. A man pushes a cylinder of mass m_1 with the help of a plank of mass m_2 as shown in fig. There is no slipping at any contact. The horizontal component of the force applied by the man is F . Find



- the acceleration of the plank and the centre of mass of the cylinder.
- the magnitudes and directions of frictional forces at contact points.

Solution From FBD, we have $f_1 + f_2 = m_1 a_1$ (1)

$$F - f_2 = m_2 a_2 \quad \text{.....(2)}$$



About centre C of cylinder,

$$(f_2 - f_1)r = \frac{m_1 r^2}{2} \alpha = \frac{m_1 r^2}{2} \left(\frac{a_1}{r} \right)$$

$$\Rightarrow f_2 - f_1 = \frac{m_1 a_1}{2} \quad \text{.....(3)}$$

As there is no slipping, $a_p = a_c$

$$\Rightarrow a_2 = \alpha(2r) = \frac{a_1}{r}(2r) = 2a_1 \quad \text{.....(4)}$$

(a) From (1), (2), (3) and (4),

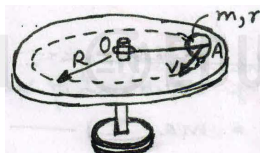
$$a_1 = \frac{4F}{3m_1 + 8m_2} \quad \text{and} \quad a_2 = \frac{8F}{3m_1 + 8m_2}$$

(b) From eqns (1) and (2),

$$f_1 = \frac{m_1 a_1}{4} = \frac{m_1 F}{3m_1 + 8m_2} \quad \text{and} \quad f_2 = \frac{3}{4} m_1 a_1 = \frac{3m_1 F}{3m_1 + 8m_2}$$

As both f_1 and f_2 are positive, they act in the direction shown in the figure.

10. A uniform sphere of mass m and radius r rolls without sliding over a horizontal plane, rotating about a horizontal axis OA. In the process, the center of the sphere moves with a velocity v along a circle of radius R . Find the kinetic energy of the sphere.



Solution The sphere will have kinetic energy due to rotation

about its horizontal axis and also due to rotation about vertical axis passing through O.

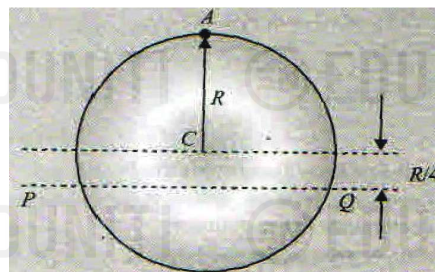
$$KE = \frac{1}{2} I_{cm} \omega_1^2 + \frac{1}{2} I_O \omega_2^2 = \frac{1}{2} \left(\frac{2}{5} m r^2 \right) \omega_1^2 + \frac{1}{2} \left(\frac{2}{5} m r^2 + m R^2 \right) \omega_2^2$$

where, $\omega_1 = v/r$ and $\omega_2 = v/R$.

On putting these values, we get

$$KE = \frac{7mv^2}{10} \left(1 + \frac{2r^2}{7R^2} \right)$$

11. A uniform circular disc has radius R and mass m . A particle, also of mass m , is fixed at point A on the edge of the disc as shown in fig. The disc can rotate freely about a fixed horizontal chord PQ that is at a distance $R/4$ from the centre C of the disc. The line AC is perpendicular to PQ. Initially, the disc is held vertical with point A at its highest position. It is then allowed to fall so that it starts rotating about PQ. Find the linear speed of the particle as it reaches its lowest position.



Solution Let the initial and final positions of the system be when point A is at highest and lowest positions respectively. From initial to final position, the distance moved by centre of mass of disc is $2(R/4) = R/2$ downwards and by particle is

$$2(R + R/4) = 5R/2 \text{ downwards.}$$

Moment of inertia of the system about PQ is

$$I_{PQ} = (I_{disc})_{PQ} + (I_{particle})_{PQ}$$

$$= \left[\frac{mR^2}{4} + m \left(\frac{R}{4} \right)^2 \right] + m \left(\frac{5R}{4} \right)^2 = \frac{15mR^2}{8}$$

By conservation of mechanical energy,

Gain in KE = Loss in PE

$$\Rightarrow \frac{1}{2} I_{PQ} \omega^2 = mg(R/2) + mg(5R/2)$$

$$\Rightarrow \frac{1}{2} \left(\frac{15mR^2}{8} \right) \omega^2 = 3mgR \quad \Rightarrow \quad \omega = 4 \sqrt{\frac{g}{5R}}$$

Therefore, the linear speed of the particle is

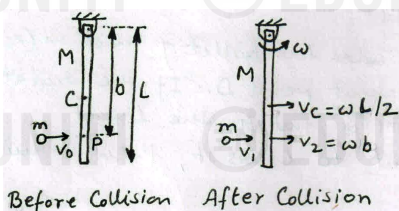
$$v = \omega \frac{5R}{4} = 4 \sqrt{\frac{g}{5R}} \times \frac{5R}{4} = \sqrt{5gR}$$

12. A ball of mass m collides elastically with a smooth hanging rod of mass M and length l .

- If $M = 3m$, find the value b for which no horizontal reaction occurs at the pivot.
- For what value of m/M , the ball falls dead just after the collision assuming $e = 1/2$

Solution

- As there is no horizontal reaction at the pivot, there is no impulse on the system in horizontal direction. By conservation of linear momentum,



$$mv_0 = mv_1 + M\omega l/2$$

$$\Rightarrow m(v_0 - v_1) = M\omega l/2 \quad \dots(1)$$

By conservation of angular momentum about the pivot,

$$mv_0 b = mv_1 b + (ML^2/3)\omega$$

$$\Rightarrow m(v_0 - v_1)b = M\omega L^2/3 \quad \dots(2)$$

From (1) and (2), $b = 2l/3$

- In this case, there may be horizontal reaction at the pivot. Now, linear momentum may not be conserved but angular momentum about the pivot will still be conserved. So, equation (2) is still valid. Here, $v_1 = 0$ and $e = 1/2$

At the point of collision,

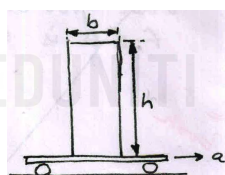
$$v_2 - v_1 = e(u_1 - u_2) \Rightarrow \omega b - 0 = \frac{1}{2}(v_0 - 0)$$

$$\Rightarrow \omega = v_0/2b$$

Putting this and $v_1 = 0$ in equation (3), we get

$$mv_0 b = \frac{ML^2}{3} \times \frac{v_0}{2b} \quad \therefore \frac{m}{M} = \frac{l^2}{6b^2}$$

13. A block of mass m , height h and width b rests on a flat car which moves horizontally with constant acceleration 'a' as shown in figure. Determine.



- the value of the acceleration at which slipping of the block on the car starts, if the coefficient of friction is μ
- the value of the acceleration at which block topples about A, assuming sufficient friction to prevent slipping and
- the shortest distance in which it can be stopped from a speed of 20 m s^{-1} with constant deceleration so that the block is not disturbed.

The following data are given

$$b = 0.2 \text{ m}, h = 0.5 \text{ m}, \mu = 0.5 \text{ and } g = 10 \text{ m s}^{-2}$$

Solution In the frame of car, pseudo force ma acts on the block

towards left. It passes through centre C of the block.

- The slipping of the block starts when static friction force is maximum which is equal to μmg

$$\therefore a_{\max} = \frac{f_{\max}}{m} = \frac{\mu mg}{m} = \mu g = 0.5 \times 10 = 5 \text{ ms}^{-2}$$

- When block is about to topple about point P, normal reaction N will pass through P.

The block topples about P when torque due to ma exceeds the torque due to mg

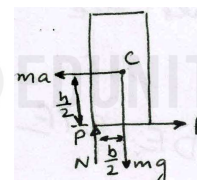
$$\text{or } ma(h/2) > mg(b/2)$$

$$\text{or } a > bg/h = 0.2 \times 10/0.5 = 4 \text{ ms}^{-2}$$

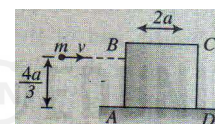
- Now, $a_{\max} = 5 \text{ ms}^{-2}$ for no slipping and $a_{\max} = 4 \text{ ms}^{-2}$ for no toppling. Hence, the maximum deceleration of car so that the block is not disturbed is 4 ms^{-2} .

$$\text{From } v^2 - u^2 = 2as, \text{ we have } 0^2 - 20^2 = 2(-4)s_{\min}$$

$$\therefore s_{\min} = 50 \text{ m}$$



14. A solid cube of wood of side $2a$ and mass M is resting on a horizontal surface. The cube is constrained to rotate about an axis passing through A and perpendicular to face ABCD. A bullet of mass m and speed v is shot at a height of $4a/3$ as shown in the figure. The bullet becomes embedded in the cube. Find the minimum value of v required to topple the cube. Assume $m \ll M$.

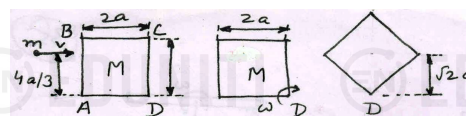


Solution During collision, angular momentum about D is conserved.

$$\Rightarrow mv(4a/3) = I_D \omega$$

$$I_D = I_{cm} + M(\sqrt{2}a)^2 = \frac{M(2a)^2}{6} + 2Ma^2 = \frac{8}{3}Ma^2$$

$$\Rightarrow mv \frac{4a}{3} = \frac{8}{3}Ma^2 \omega \Rightarrow \omega = \frac{mv}{2Ma}$$



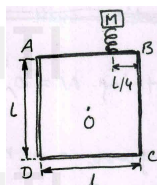
The cube will topple if it manages to rotate by $\pi/4$. Applying energy conservation after collision, we have

$$\frac{1}{2}I_D \omega^2 = Mg(\sqrt{2}a - a)$$

$$\Rightarrow \frac{1}{2} \left(\frac{8}{3}Ma^2 \right) \left(\frac{mv}{2Ma} \right)^2 = Mga(\sqrt{2} - 1)$$

$$\therefore v = \frac{M}{m} \sqrt{3ga(\sqrt{2} - 1)}$$

15. A square frame is formed by four rods, each of length $l = 60$ cm. Mass of two rods AB and BC is $m_1 = 25/18$ kg each while that of rods AD and CD is $m_2 = 2$ kg each. The frame is free to rotate about a fixed horizontal axis passing through its geometric centre O as shown in figure. A spring is placed on the rod AB at a distance $l/4 = 15$ cm from B. The spring is held vertical and a block is placed on upper end of the spring so that rod AB is horizontal.



- Calculate mass M of the block
- If the spring is initially compressed by connecting a thread between its ends and energy stored in it is 76.5 J, calculate velocity with which block bounces up when the thread is burnt.

Solution

- As the system is in equilibrium, the net torque about O is zero. There is no torque due to rods AB and CD. The clockwise torque due to M and rod BC is equal to the anticlockwise torque due to rod AD.

$$\Rightarrow Mgl/4 + m_1 gl/2 = m_2 gl/2$$

$$\therefore M = 2(m_2 - m_1) = 2(2 - 25/18) = \boxed{11/9 \text{ kg}}$$

- Moment of inertia of four rods about O is

$$I = 2 \left[\frac{m_1 l^2}{12} + m_1 \left(\frac{l}{12} \right)^2 \right] + 2 \left[\frac{m_2 l^2}{12} + m_2 \left(\frac{l}{2} \right)^2 \right]$$

$$= \frac{2}{3} (m_1 + m_2) l^2 = \frac{2}{3} \left(\frac{25}{18} + 2 \right) (0.6)^2 = \frac{61}{75} \text{ kg m}^2$$

Let v be the velocity by which the block bounces up when the thread is burnt. Then, by conservation of mechanical energy,

$$\frac{1}{2} Mv^2 + \frac{1}{2} I\omega^2 = 76.5 \quad \dots(1)$$

and by conservation of angular momentum about O,

$$Mvl/4 = I\omega \Rightarrow \omega = Mvl/4I$$

Putting this in eqn (1), we get

$$Mv^2 + I \left(\frac{Mvl}{4I} \right)^2 = 2 \times 76.5$$

$$\Rightarrow Mv^2 \left(1 + \frac{Ml^2}{16I} \right) = 153$$

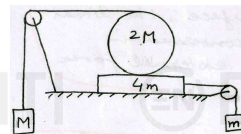
$$\Rightarrow \frac{11}{9} v^2 \left[1 + \frac{(11/9) 0.6^2}{16 (61/75)} \right] = 153$$

$$\therefore \boxed{v = 11 \text{ ms}^{-1}}$$

PARAGRAPH TYPE QUESTIONS

Paragraph for Question Nos. 16 to 18

Figure shows a uniform smooth solid cylinder A of radius 4 m rolling without slipping on the 8 kg plank which, in turn is supported by a fixed smooth surface. Block B is known to accelerate down with 6 m/s^2 and block C moves down with acceleration 2 m/s^2



- What is the angular acceleration of the cylinder
 - $4/5 \text{ rad s}^{-2}$
 - $6/5 \text{ rad s}^{-2}$
 - 2 rad s^{-2}
 - 1 rad s^{-2}
- What is the ratio of the mass of the cylinder to the mass of block B?
 - 1
 - 2
 - 3
 - 4
- If unwrapped length of the thread between the cylinder and block B is 20 m at the beginning, when the system was released from rest, what would it be 2 s later?
 - 28 m
 - 30 m
 - 22 m
 - 32.5 m

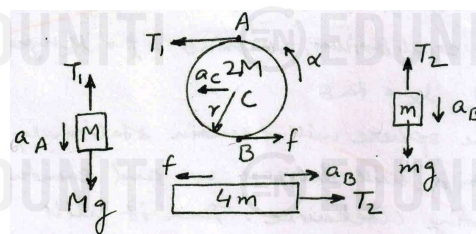
Solution

- From FBD,

$$Ma_A = Mg - T_1 \quad \dots(1)$$

$$ma_B = mg - T_2 \quad \dots(2)$$

$$4m a_B = T_2 - f \quad \dots(3)$$



$$a_C = (a_A - a_B)/2$$

$$\Rightarrow 2M \left(\frac{a_A - a_B}{2} \right) = T_1 - f \quad \dots(4)$$

$$\alpha = \frac{a_A + a_B}{2r}$$

$$\text{About C, } \left(\frac{2Mr^2}{2} \right) \left(\frac{a_A + a_B}{2r} \right) = (T_1 + f)r$$

$$\Rightarrow \frac{M}{2} (a_A + a_B) = T_1 + f \quad \dots(5)$$

$$\text{From (2) and (3), } 5ma_B = mg - f \quad \dots(6)$$

$$\text{From (1) and (4), } M(2a_A - a_B) = Mg - f \quad \dots(7)$$

From (1) and (5), $M\left(\frac{3a_A}{2} + \frac{a_B}{2}\right) = Mg + f$ (8)

From (7) and (8),

$$\frac{M}{2}(7a_A - a_B) = 2Mg \Rightarrow 7a_A - a_B = 4g \text{(9)}$$

From (6) and (8), $\frac{3}{2}Ma_A + \left(\frac{M}{2} + 5m\right)a_B = (M + m)g$
 $\Rightarrow 3Ma_A + (M + 10m)a_B = 2(M + m)g$ (10)

From (9) and (10), $a_A = 0.6g$ and $a_B = 0.2g$

$$\therefore \alpha = \frac{a_A + a_B}{2r} = \frac{0.6g + 0.2g}{2 \times 4} = 1 \text{ rad/s}^2$$

17. (c) $\frac{a_A}{a_B} = \frac{0.6g}{0.2g} = 3$

18. (a) $a_C = (a_A - a_B)/2 = (0.6g - 0.2g)/2 = 2 \text{ m/s}^2$

The acceleration of top part of cylinder w.r.t. its COM is

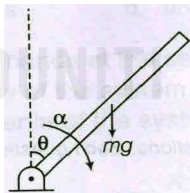
$$a_{AC} = a_A - a_C = 6 - 2 = 4 \text{ m/s}^2$$

Length of unwrapped thread after time 2sec is

$$20 + \frac{1}{2}a_{AC} \times 2^2 = 28m.$$

Paragraph for Question Nos. 19 to 21

A uniform rod of length L and mass m is pivoted freely at one end as shown in the figure. Answer the following questions.



19. The angular acceleration of the rod when it is at an angle θ to the vertical is

- (a) $\frac{2g \sin \theta}{3L}$ (b) $\frac{g \sin \theta}{3L}$
 (c) $\frac{4g \sin \theta}{3L}$ (d) $\frac{3g \sin \theta}{2L}$

20. Assuming the rod to start from the vertical position, the angular velocity as a function of θ is

- (a) $\omega = \sqrt{\frac{3g}{L}(1 - \sin \theta)}$ (b) $\omega = \sqrt{\frac{2g}{L}(1 - \sin \theta)}$
 (c) $\omega = \sqrt{\frac{3g}{L}(1 - \cos \theta)}$ (d) $\omega = \sqrt{\frac{2g}{L}(1 - \cos \theta)}$

21. The tangential linear acceleration of the free end when the rod is horizontal, is

- (a) $2g/3$ (b) $3g/2$
 (c) $4g/3$ (d) $g/2$

Solution

19. (d) From $\tau = I\alpha$,

$$mg \frac{L}{2} \sin \theta = \frac{mL^2}{3} \alpha \therefore \alpha = \frac{3g \sin \theta}{2L}$$

20. (c) $\alpha = \omega \frac{d\omega}{d\theta} = \frac{3g \sin \theta}{2L} \Rightarrow \int_0^\omega \omega d\omega = \frac{3g}{2L} \int_0^\theta \sin \theta d\theta$

$$\Rightarrow \frac{\omega^2}{2} = \frac{3g}{2L} [-\cos \theta]_0^\theta \therefore \omega = \sqrt{\frac{3g}{L}(1 - \cos \theta)}$$

Alternatively, by energy conservation,

Gain in KE = Loss in PE

$$\Rightarrow \frac{1}{2} \left(\frac{mL^2}{3} \right) \omega^2 = mg \frac{L}{2} (1 - \cos \theta)$$

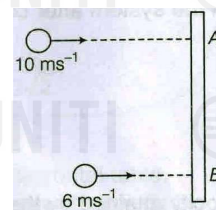
$$\therefore \omega = \sqrt{\frac{3g}{L}(1 - \cos \theta)}$$

21. (b) When the rod is horizontal, $\theta = 90^\circ$

$$\Rightarrow \alpha = 3g/2L \therefore a_t = \alpha L = 3g/2$$

Paragraph for Question Nos. 22 to 24

A thin uniform bar lies on a frictionless horizontal surface and is free to move in any way on the surface. Its mass is 0.16 kg and length is $\sqrt{3}m$. Two particles, each of mass 0.08 kg are moving on the same surface and towards the bar in a direction perpendicular to the bar, one with a velocity 10 ms^{-1} and the other with 6 ms^{-1} as shown in the figure. The first particle strikes the bar at point A and the other at point B. Each of A and B is at a distance of $0.5 m$ from the centre of the bar. The particles strike the bar at the same instant of time and stick to the bar after collision.



22. The velocity of centre of mass of the system just after impact (in ms^{-1}) is

- (a) 1 (b) 2
 (c) 3 (d) 4

23. The angular velocity of the system just after impact (in rad/s) is

- (a) 8 (b) 4
 (c) 2 (d) 1

24. The loss in kinetic energy of the system in the above collision process is

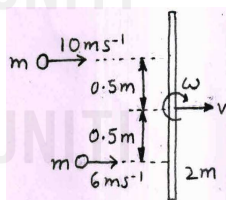
- (a) 2.72 J (b) 1.72 J
 (c) 3.36 J (d) 4.36 J

Solution

22. (d) By cons. of linear momentum,

$$m \times 10 + m \times 6 = (m + m + 2m)v$$

$$\therefore v = 4 \text{ ms}^{-1}$$



$$23. (c) m \times 10 \times 0.5 - m \times 6 \times 0.5 = \left[\frac{2m(\sqrt{3})^2}{12} + 2 \times m(0.5)^2 \right] \omega$$

$$\Rightarrow 2m = m\omega$$

$$\therefore \omega = 2 \text{ rad s}^{-1}$$

24. (a) $m = 0.08 \text{ kg}$,

$$I = \frac{2m(\sqrt{3})^2}{12} + 2 \times m(0.5)^2 = m = 0.08 \text{ kg m}^2$$

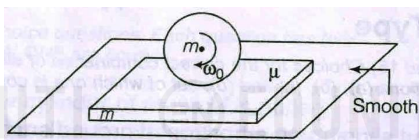
Loss in KE

$$= \left[\frac{1}{2} m \times 10^2 + \frac{1}{2} m \times 6^2 \right] - \left[\frac{1}{2} (4m)v^2 + \frac{1}{2} I \omega^2 \right]$$

$$= 68m - \left[2m \times 4^2 + \frac{1}{2} \times m \times 2^2 \right] = 34m = 34 \times 0.08 = 2.72 \text{ J}$$

Paragraph for Question Nos. 25 to 27

A long horizontal plank of mass m is lying on a smooth horizontal surface. A sphere of same mass m and radius R is spun about its own axis with angular velocity ω_0 and gently placed on the plank. The coefficient of friction between the plank and the sphere is μ . After some time, pure rolling of the sphere on the plank starts.



25. The time after which pure rolling begins is

(a) $\frac{R\omega_0}{9\mu g}$

(b) $\frac{2R\omega_0}{9\mu g}$

(c) $\frac{R\omega_0}{3\mu g}$

(d) $\frac{2R\omega_0}{\mu g}$

26. The velocity of the sphere after pure rolling begins is

(a) $\frac{2}{9} R\omega_0$

(b) $\frac{R\omega_0}{9}$

(c) $\frac{R\omega_0}{3}$

(d) $\frac{2}{3} R\omega_0$

27. The displacement of the plank till the sphere starts pure rolling is

(a) $\frac{R^2\omega_0^2}{81\mu g}$

(b) $\frac{2}{27} \left[\frac{R^2\omega_0^2}{\mu g} \right]$

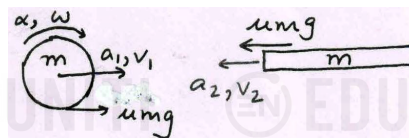
(c) $\frac{4}{81} \left[\frac{R^2\omega_0^2}{\mu g} \right]$

(d) $\frac{2}{81} \left[\frac{R^2\omega_0^2}{\mu g} \right]$

Solution

$$25. (b) \text{ From FBD, } a_1 = \frac{\mu mg}{m} = \mu g, \quad a_2 = \frac{\mu mg}{m} = \mu g,$$

$$\alpha = \frac{-\mu mg}{2mR^2/5} = -\frac{5\mu g}{2R}$$



When pure rolling starts, $v_1 + v_2 = \omega R$

$$\Rightarrow a_1 t + a_2 t = (\omega_0 + \alpha t) R$$

$$\therefore t = \frac{\omega_0 R}{a_1 + a_2 - \alpha R} = \frac{\omega_0 R}{\mu g + \mu g - 5\mu g/2} = \frac{2\omega_0 R}{9\mu g}$$

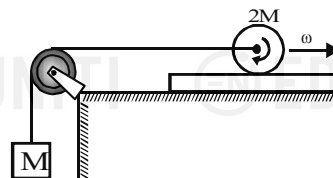
$$26. (a) v_1 = a_1 t = \mu g \times \frac{2\omega_0 R}{9\mu g} = \frac{2}{9} \omega_0 R$$

27. (d) Displacement of plank,

$$s = \frac{1}{2} a_2 t^2 = \frac{1}{2} \mu g \left(\frac{2}{9} \frac{\omega_0 R}{\mu g} \right)^2 = \frac{2}{81} \left(\frac{\omega_0^2 R^2}{\mu g} \right)$$

Paragraph for Questions Nos. 28 to 30

A disc of mass $2M$ and radius R is placed on a fixed plank (rough) of length L . The coefficient of friction between the plank and disc is $\mu = 0.5$. A light string is connected to centre of disc and passes over a smooth light pulley and connected to a block of mass M as shown in the figure. Now the disc is given an angular velocity ω_0 in clockwise direction and is gently placed on the plank at its right end. Consider this instant as $t = 0$. Based on above information, answer the following questions



28. Mark the correct statement w.r.t. motion of block and disc.

(a) The block remains at rest for some time, $t > 0$.

(b) The block starts accelerating just after placing of disc on plank.

(c) The disc is performing pure rotational motion for some time $t > 0$

(d) Both (a) and (c) are correct.

29. Time t_0 upto which the block remains stationary is

(a) $\frac{\omega_0 R}{g}$

(b) $\frac{4\omega_0 R}{g}$

(c) Zero

(d) Question is irrelevant



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ROTATION

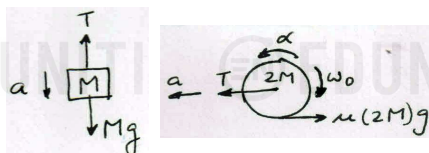
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30. Time t_1 at which the disc will cross the other end of the plank is

(a) $\sqrt{\frac{8L}{g}}$ (b) $\frac{\omega_0 R}{g} + \sqrt{\frac{8L}{g}}$
 (c) $\sqrt{\frac{8L}{g}} + \frac{4\omega_0 R}{g}$ (d) $\sqrt{\frac{\omega_0 R}{g}} + \sqrt{\frac{4L}{g}}$

Solution

28. (d) As the disc is rotating clockwise, friction is initially kinetic and direction is towards right.



From FBD, $Ma = Mg - T$ (1)

$2Ma = T - \mu(2M)g \Rightarrow 2Ma = T - Mg$ (2)

About centre of disc, $\frac{2(M)R^2}{2}\alpha = \mu(2M)gR$ (3)

From (1) and (2), $a = 0$.

Therefore, the block remains at rest till angular velocity of the disc becomes zero. Till this time, the disc is performing pure rotational motion.

29. (a) From (3), $\alpha = g/R \Rightarrow 0 - (-\omega_0) = \alpha t$

$\therefore t_0 = \frac{\omega_0}{\alpha} = \frac{\omega_0}{g/R} = \frac{\omega_0 R}{g}$

30. (b) After the disc stops rotating, rolling starts. The disc starts moving towards left and the friction becomes static. Let it be f . Now,

$Ma = Mg - T, 2Ma = T - f$ and $\frac{(2M)R^2}{2}\left(\frac{a}{R}\right) = fR$

On solving, we get $a = g/4$.

Time taken to reach the other end of the plank after the disc starts moving is

$t = \sqrt{\frac{2L}{a}} = \sqrt{\frac{2L}{g/4}} = \sqrt{\frac{8L}{g}}$

Total time, $t_1 = t_0 + t = \frac{\omega_0 R}{g} + \sqrt{\frac{8L}{g}}$